MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, TTL, LOW POWER, EXCLUSIVE-OR GATES, MONOLITHIC SILICON

This limited coordination Military specification has been prepared by the Air Force Space and Missile Systems Organization based upon currently available technical information, but it has not been approved for promulgation as a coordinated revision of Military Specification MIL-M-38510/26(USAF). It is subject to modification. However, pending its promulgation as a coordinated Military specification, it may be used in procurement.

1. SCOPE

1.1 Scope. This specification covers the requirements for monolithic silicon, TTL, low power, quadruple, 2-input, exclusive-or gate microcircuits, for use in Air Force missile and space systems.

1.2 Part number. The complete part number shall be as shown in the following example:

M38510	/026	01	S	T	D
Military designator	Detail specification	Device type (1.2.1)	Device class (1.2.2)	Case outline (1.2.3)	Lead finish (3.3)

1.2.1 Device type. The device type shall be as shown in the following:

Device type	Circuit	Similar commercial type
01	Quad, 2-input exclusive-or gates	54L86

1.2.2 Device class. The device class shall be Class S as defined in MIL-M-0038510.

1.2.3 Case outline. The case outline shall be as specified in figure 10 of MIL-M-0038510 and shall be identified as outline T.

1.3 Absolute maximum ratings.

Supply voltage range	0 to 8.0 Vdc
Input voltage range	0 to 5.5 Vdc
Storage temperature range	-65°C to 150°C
Lead temperature (soldering, 10 seconds)	300°C

1.4 Recommended operating conditions.

Ambient operating temperature range	-55°C to 125°C
Supply voltage	4.5 Vdc minimum to 5.5 Vdc maximum
Minimum high level input voltage	2.0 Vdc
Maximum low level input voltage	0.7 Vdc

Source: https://assist.dla.mil -- Downloaded: 202610-21T12:57Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 The following document, of the issue in effect on date of invitation for bids or request for proposal, forms a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-0038510 - Microcircuits, General Specification for.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-0038510 and as specified herein.

3.2 Design, construction, weight and physical dimensions. The design, construction, weight and physical dimensions shall be as specified in MIL-M-0038510 and herein.

3.2.1 Logic diagram and terminal connections. The logic diagram and terminal connections shall be as specified on figure 1.

3.2.2 Truth tables and logic equations. The truth tables and logic equations shall be as specified on figure 2.

3.2.3 Schematic circuit. The schematic circuit shall be as specified on figure 3.

3.2.4 Case outline. The case outline shall be as specified in figure 10 of MIL-M-0038510 and shall be identified as case outline T.

3.2.5 Weight. The maximum weight shall be 1 gram.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-0038510.

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in tables I and II.

3.4.1 Delta drift limits. The delta drift limits shall be as specified in table I.

3.5 Screening. Screening shall be in accordance with table I of MIL-M-0038510 and as specified herein.

3.5.1 Interim electrical parameters. The interim electrical parameters shall consist of all the electrical characteristics tests as specified in table I. Each test shall be performed go, no-go at -55°C and 125°C. Each test shall also be read and recorded at 25°C. For all tests, the devices shall meet the limits specified in table I.

3.5.2 Final electrical tests. The final electrical tests shall consist of all the electrical characteristics tests specified in table I. Each test shall be read and recorded at -55°C, 25°C and 125°C. For all tests, the devices shall meet the limits specified in table I.

3.5.2.1 Delta computation. Delta computations shall be performed for those parameters in table I which are marked with an asterisk. All devices shall meet the applicable delta limits specified in table I. (See section 6 in MIL-M-0038510).

3.5.3 Switching characteristics tests. The switching characteristics tests specified in table II and figure 4 shall be performed at 25°C. All devices shall meet the limits specified in table II.

Source: https://assist.dla.mil -- Downloaded: 202610-21T12:57Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical characteristics tests.

Test No. 1/2/	Characteristics	Symbol	Conditions	Limits 3/		Delta (Δ) 1/	Units
				Min	Max		
1	Input voltage required to ensure a high level input (each input)	V_{IH}	V_{CC} = 4.5 V	2	-	-	V
2	Input voltage required to ensure a low level input (each input)	V_{IL}	V_{CC} = 4.5 V	-	0.7	-	V
3*	High level output voltage	V_{OL}	V_{CC} = 4.5 V, V_{IH} = 2 V V_{IL} = 0.7 V, I_{OH} = -100 µA	2.4	-	±.12	V
4*	Low level output voltage	V_{OL}	V_{CC} = 4.5 V, V_{IH} = 2 V V_{IL} = 0.7 V, I_{OL} = 2 mA	-	0.3	±.03	V
5*	High level input current (each input)	I_{IH}	V_{CC} = 5.5 V, V_{IH} = 2.4 V	-	20	10X or 2 µA	µA
			V_{CC} = 5.5 V, V_{IH} = 5.5 V	-	200	N/A	µA
6*	Low level input current (each input)	I_{IL}	V_{CC} = 5.5 V, V_{IL} = 0.3 V	-	-0.36	±.02	mA
7	Short circuit output current	I_{OS}	V_{CC} = 5.5 V, V_{IH} = 4.5 V V_{IL} = 0 V	-3	-15	-	mA
8	Supply current	I_{CCL}	V_{CC} = 5.5 V, V_{IH} = 4.5 V	-	6.68	-	mA
9	Supply current	I_{CCH}	V_{CC} = 5.5 V, V_{IH} = 4.5 V V_{IL} = 0 V	-	4.4	-	mA

1/ Delta computation shall be performed on parameters marked with an asterisk (*) with data read and recorded ("10X" denotes "10 times the initial measurement"). Where two delta values are indicated for one parameter, the greater value shall apply.
2/ Test sequence is optional with the manufacturer.
3/ Unless otherwise specified, the limits apply at T_A = -55°C to +125°C.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:57Z
Check the source to verify that this is the current version before use.

TABLE II. Switching characteristics.

Characteristic	Symbol	Conditions	Test Figure	Limits Min	Limits Max	Units
Propagation delay time to low level output (other input low)	t_{PHL_1}	C_L = 50 pF, R_L = 4 kΩ ±5%, V_{CC} = 5 V, T_A = 25°C, N = 10	4	-	150	ns
Propagation delay time to high level output (other input low)	t_{PLH_1}			-	150	ns
Propagation delay time to low level output (other input high)	t_{PHL_2}			-	60	ns
Propagation delay time to high level output (other input high)	t_{PLH_2}			-	90	ns

Source: https://assist.dla.mil -- Downloaded: 2026-04-21T12:57Z
Check the source to verify that this is the current version before use.

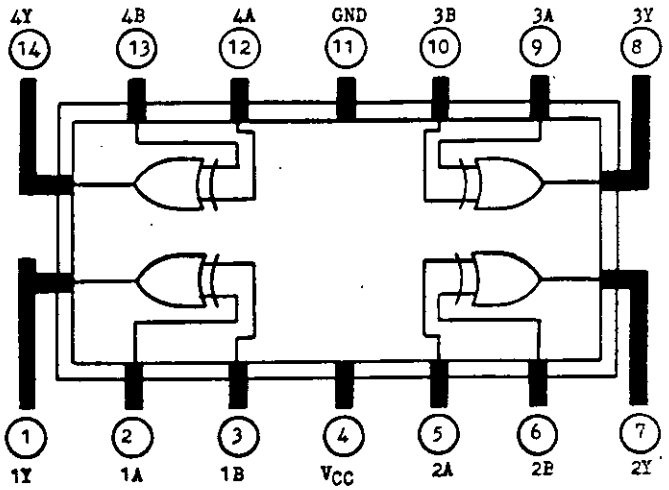


FIGURE 1. Logic diagram and terminal connections.

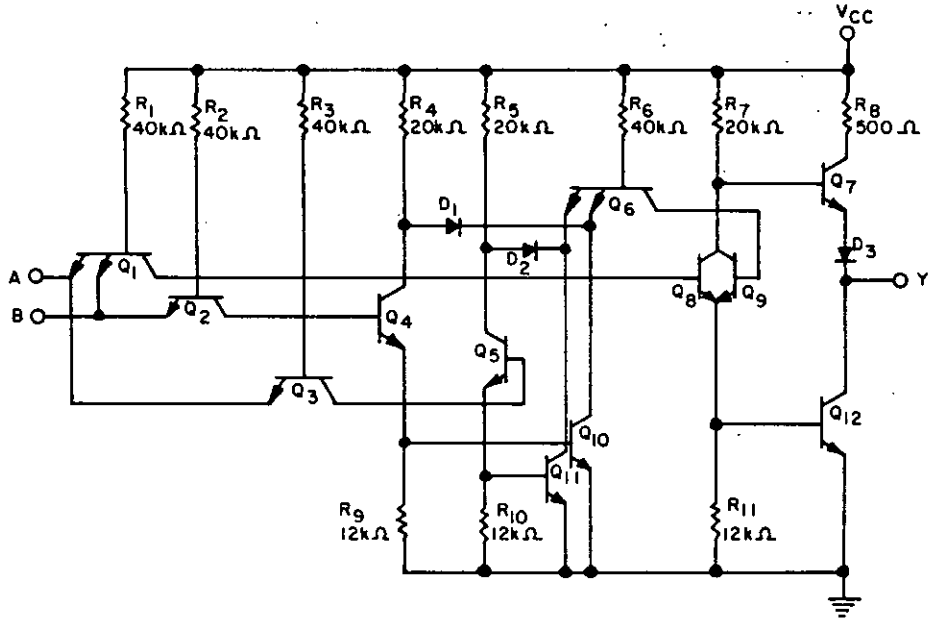
TRUTH TABLE (LOGIC)

INPUTS		OUTPUT
A	B	Y
L	L	L
L	H	H
H	L	H
H	H	L

Positive logic: $Y = A \oplus B$

FIGURE 2. Truth table and logic equation.

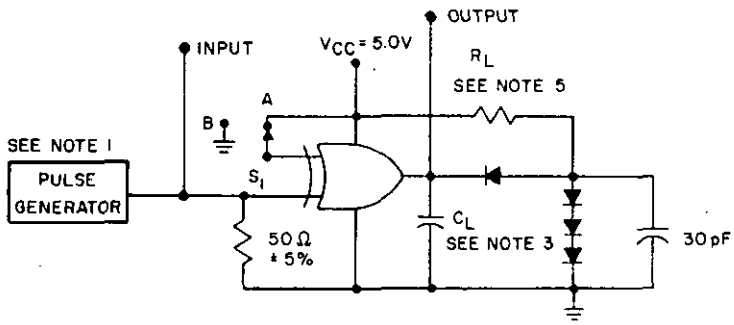
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:57Z
Check the source to verify that this is the current version before use.

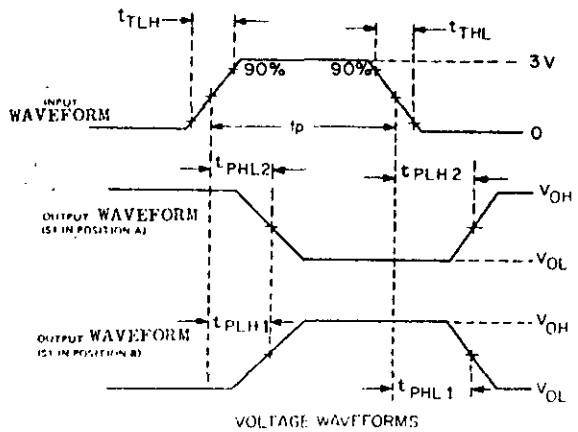


NOTE: All resistance values shown are nominal.

FIGURE 3. Schematic circuit.

Source: https://assist.dla.mil -- Downloaded: 202610-21T12:57Z
Check the source to verify that this is the current version before use.





VOLTAGE WAVEFORMS

NOTES:
1. The generator has the following characteristics: V_{gen} = 3 V, $t_{THL} = t_{TLH} \leq$ 60 ns, t_p = .5 μs, PRR ≤ 1 MHz, $Z_{out} \approx 50\ \Omega$.
2. All diodes are 1N916.
3. C_L = 50 pF minimum including scope probe, wiring, and stray capacitance, without package in test fixture.
4. Each gate tested separately.
5. R_L = 4 kΩ ±5%.
6. Propagation delay = $\frac{t_{PHL} + t_{PLH}}{2}$

FIGURE 4. Switching test circuit and waveforms.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:57Z
Check the source to verify that this is the current version before use.

3.6 Extended life tests. Extended life tests shall be performed in accordance with MIL-M-0038510.

3.7 Marking. Marking shall be in accordance with MIL-M-0038510 and 1.2 herein.

4. PRODUCT ASSURANCE PROVISIONS

4.1 Product assurance provisions. Product assurance shall be in accordance with MIL-M-0038510.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery. Preparation for delivery shall be as specified in MIL-M-0038510.

6. NOTES

6.1 Notes. Notes are as specified in MIL-M-0038510.

Custodian:
Air Force - 19

Review activities:
Air Force - 17, 19
DSA - ES

Preparing activity:
Air Force - 19

Agent:
DSA - ES

(Project 5962-F088-10)

☆ U. S. GOVERNMENT PRINTING OFFICE: 1974-713-155/6158

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:57Z
Check the source to verify that this is the current version before use.